Growth and Differentiation

08/06/2026

Key Words: Differentiation, adult stem cells, cloning, cuttings, plantlets, elongate, meristem

Starter: Let's do a quiz on what we learnt last lesson. 1→ 10

Catalysts and Enzymes

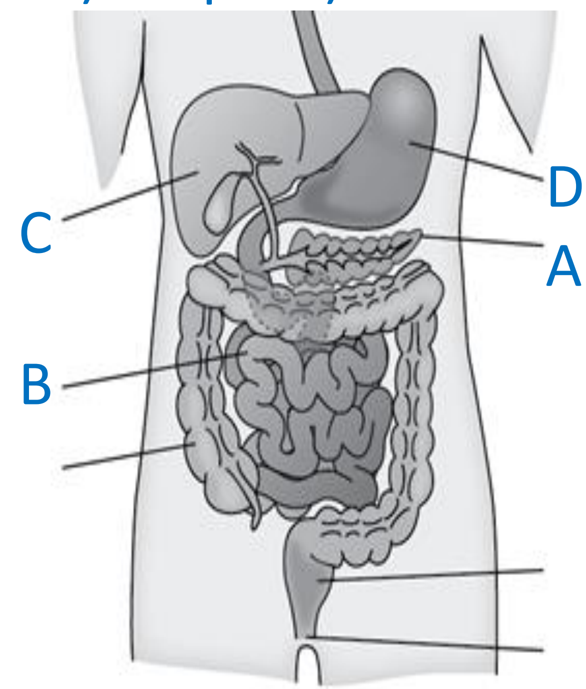
08/06/2026

Keywords: Substrate, product, active site, lock and key, enzyme, catalyst, metabolism

<u>Starter</u>: Let's do a quiz on what we learnt last lesson. 1→ 10

Quiz 4: Food Tests

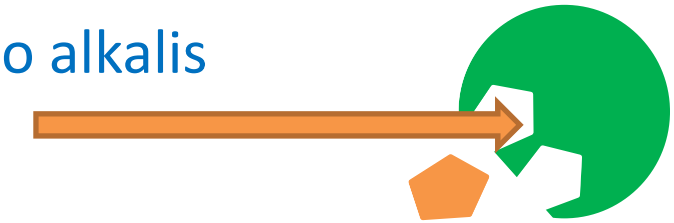
1. Which test tests for starch a) Biuret b) Iodine c) Benedict's
2. Which colour is a positive test for starch a) Yellow b) Purple c) Black
3. What is the test for sugars a) Biuret b) Iodine c) Benedict's
4. Which colour is a positive test for starch a) Yellow b) Purple c) Black
5. What is the test for protein a) Biuret b) Iodine c) Benedict's
6. Which colour is a positive test for starch a) Yellow b) Purple c) Black
7. What organ is part C
a) Stomach b) liver c) Pancreas
8. Which part produces hydrochloric acid
9. Diffusion is the movement of ________ from an area of ______ concentration to ______ concentration until E____________
10. Active transports requires energy. Which cell structure supplies energy for the cell?



Quiz 5: Enzymes

1. Enzymes are made of a) catalysts b) Carbohydrates c) proteins

2. Proteins are made of a) Amino units b) Amino alkalis c) Amino acids d) Amino nitrates



3. A catalyst is a substance that speeds up | slows down a reaction

4. Which is an enzyme that breaks down carbohydrates
a) amylose b) amylase c) amylate d) amyluse

5. Enzymes have a specific a) nature b) shape c) direction

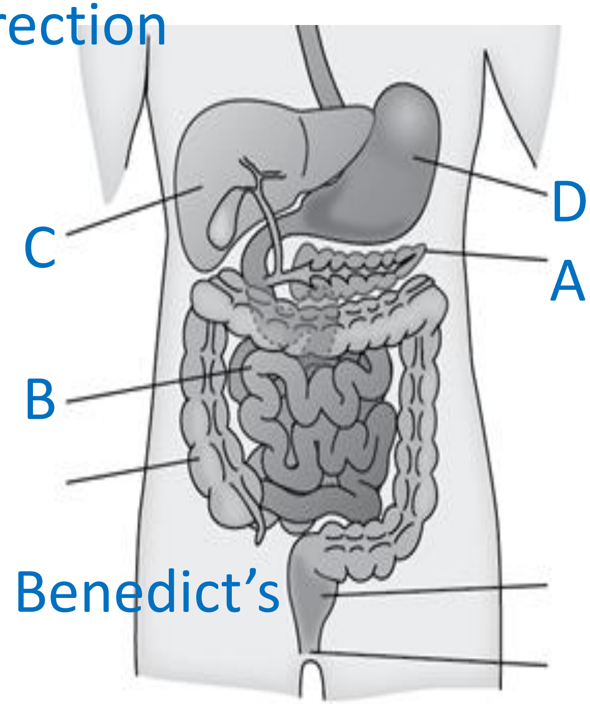
6. Order these smallest to biggest
Organ, cell, tissue, organ system

7. What organ is part B

8. Which part produces hydrochloric acid

9. What is the test for sugars a) Biuret b) Iodine c) Benedict's

10. Which test tests for starch a) Biuret b) Iodine c) Benedict's



Catalysts and Enzymes

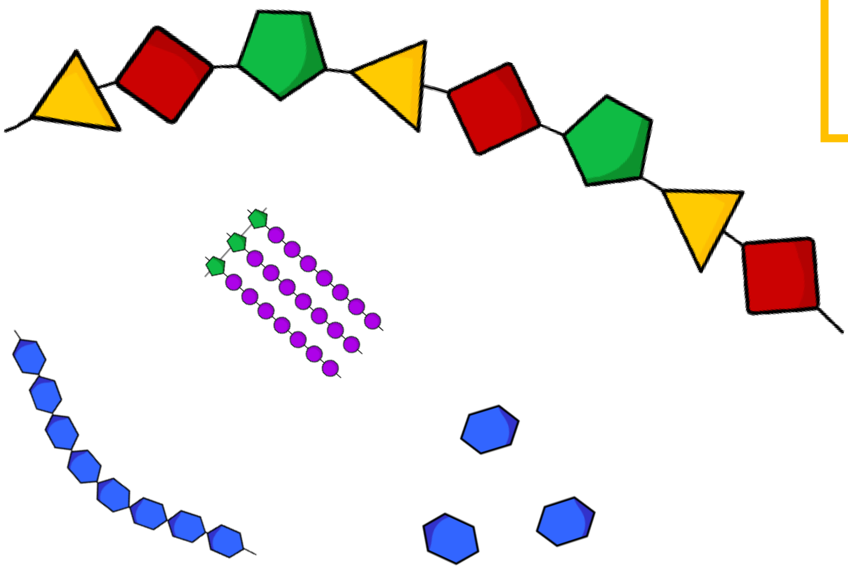
08/06/2026

Keywords: Substrate, product, active site, lock and key, enzyme, catalyst, metabolism

Starter: Answer these three review questions below.

Ext: Name three specialised cells and their function.

1. What are the three main nutrients?



2. Write down a use for each nutrient.

3. Name the reagent for each nutrient test and its positive result.

Catalysts and Enzymes

08/06/2026

Keywords: Substrate, product, active site, lock and key, enzyme, catalyst, metabolism

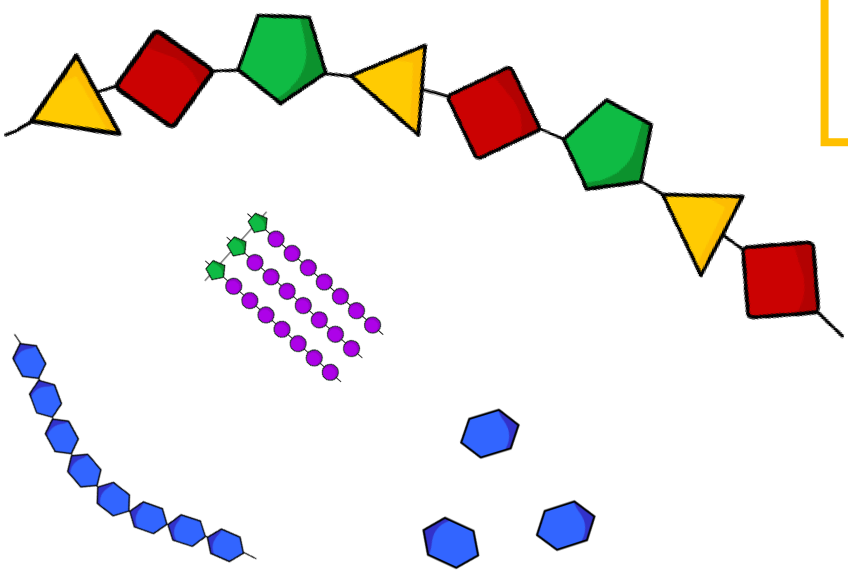
DO NOW: Answer these three review questions below.

H/W: Search " Factors affecting enzyme activity BBC Bitesize"
Make notes on page 2 and 3

1. What are the three main nutrients?

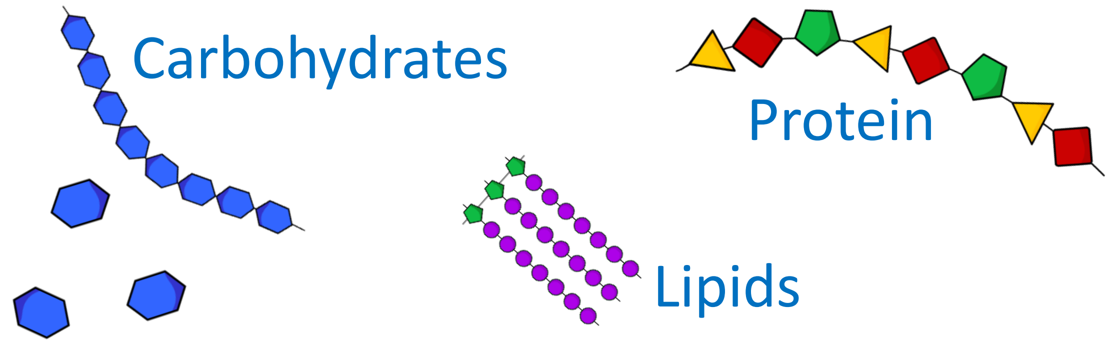
2. Write down a use for each nutrient.

3. Name the reagent for each nutrient test and its positive result.



Digestion

1. What are the three main nutrients?



2. Write down a use for each nutrient

Carbohydrates → A fuel source for respiration

Protein → Makes cell/tissue structures, enzymes

Lipid → Efficient fuel source, makes hormones

3. Name the reagent for each nutrient test and its positive result

Carbohydrates → iodine → Turns black

Protein → Biuret → purple

Sugars → Benedict's → yellow

Lipids → ethanol + water → emulsion cloudiness

LOb: Understand how enzymes help digestion.

Keywords: Substrate, product, active site, denature, pH, lock and key, enzyme, catalyst, metabolism

Success Criteria:

- Recall the definition of catalyst
- Explain how enzymes work using the lock and key method
- Explain what the metabolism of the body involves

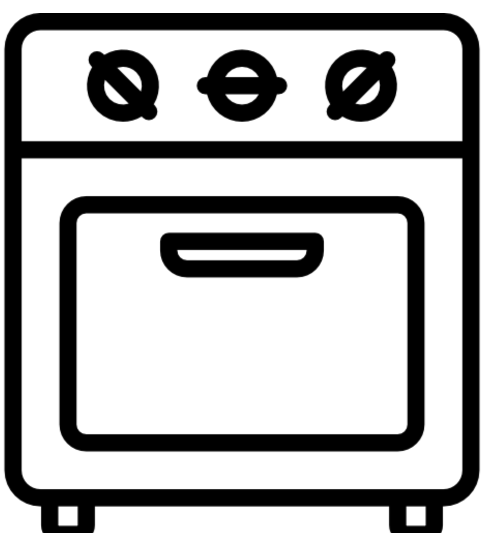
Did you know… most enzymes are named with 'ase' at the end. However, enzymes discovered longer ago are not always with an 'ase' e.g. Pepsin or trypsin.

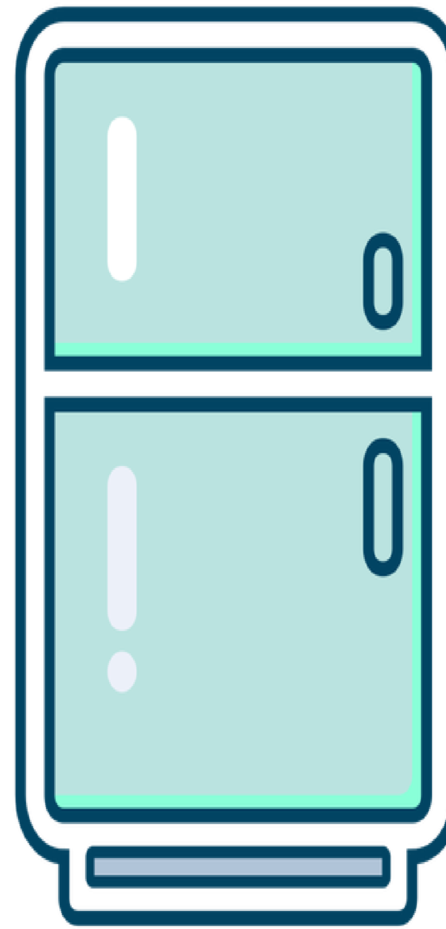
Chemical reactions are being controlled all the time.

Sometimes people use chemicals called **catalysts** to control a reaction.

A fridge slows down the chemical reactions to keep the food fresh for longer

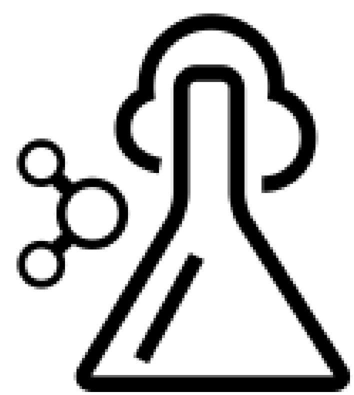
The oven speeds up the chemical reaction when you cook





Catalyst

A substance that speeds up a chemical reaction without getting used up itself.



Our bodies control chemical reactions in our cells using enzymes which are biological catalysts. Without enzymes reactions would be too slow and we would die.

FACT

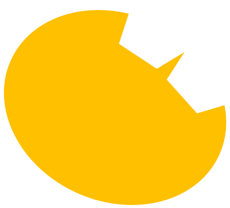
Enzymes are not alive

FACT

They don't need energy

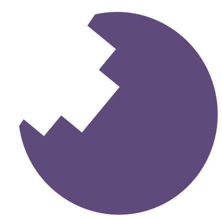
FACT

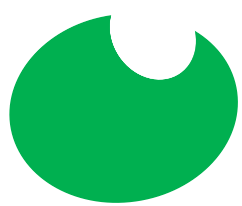
Enzymes have an active site with a specific shape



FACT

Enzymes are large proteins



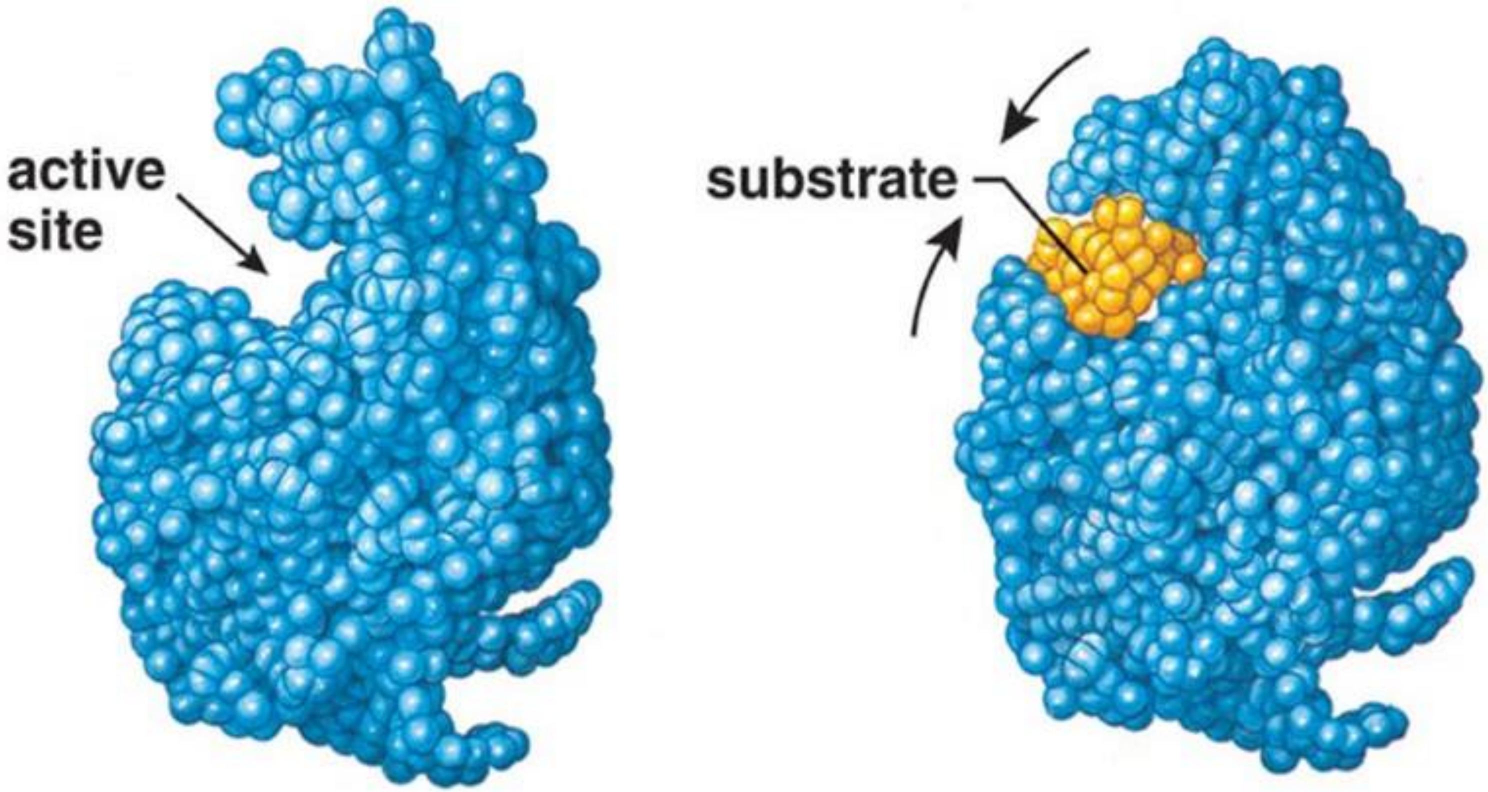


FACT

Enzymes react with a substrate

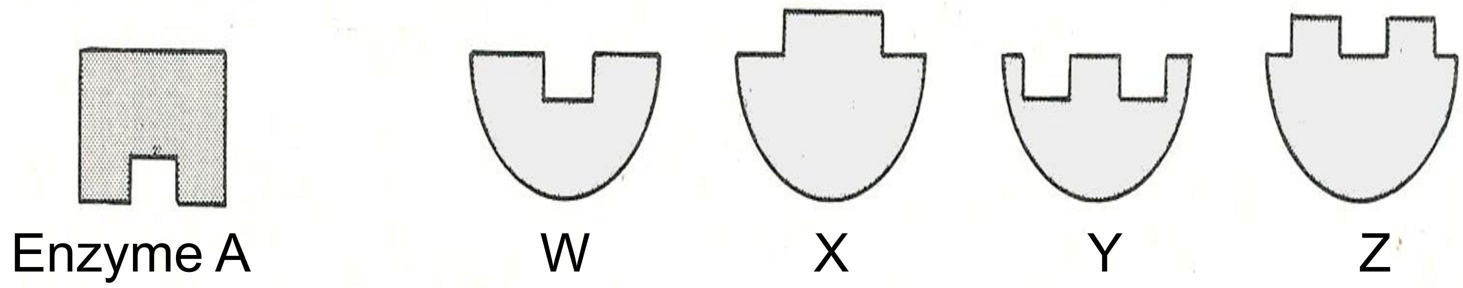
Computer generated image of an enzyme

What do you think happens once it has reacted?



Review

1. Which substrate will be acted on by enzyme A? [1 mark]



2. Complete the paragraph fill. [6]

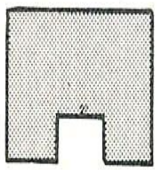
All cells contain YMZENES which are BOIGOLOIALC cTALYIASST They cTALAYES chemical reactions. The YMZENES do not get SEU up allowing them to take part in further reactions.

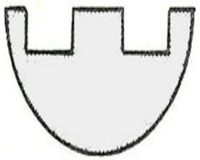
3. Other than using a biological catalyst how else can we control the speed of a reaction? [2]

4. Why is it an advantage for our bodies to use enzymes. [2]

Review

1. Which substrate will be acted on by enzyme A?





2. Complete the paragraph fill.

All cells contain enzymes which are biological catalysts. They speed up chemical reactions. The enzymes do not get used up allowing them to take part in further reactions.

3. Other than using a biological catalyst how else can be control the speed of a reaction? Ovens and fridges

4. Why is it an advantage for our bodies to use enzymes.
They don't get used up [1] They don't require energy to work [1] They speed up reactions [1]

LOb: Understand how enzymes help digestion.

Keywords: Substrate, product, active site, denature, pH, lock and key, enzyme, catalyst

Success Criteria:

- ✓ Recall names of enzymes and where they are made and work
- ➢ Observe the reaction of amylase with starch
- ➢ Explain the reaction of amylase with starch

Did you know… most enzymes are named with 'ase' at the end. However, enzymes discovered longer ago are not always with an 'ase' e.g. Pepsin or trypsin.

Activity: Watch the video describing how enzymes work.

WATCH AND

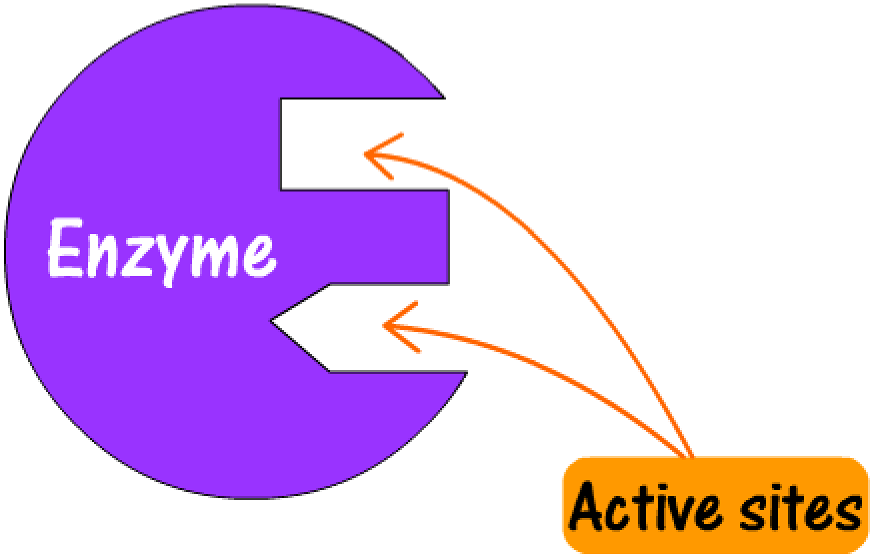
LEARN

AFTERALL KNOWLEDGE IS

POWER!

How does an enzyme work?

Enzymes are biological catalysts.
They speed up chemical reactions within the cell.

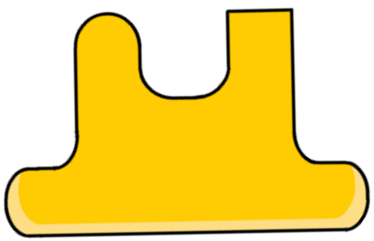


Enzymes are found in
the cells of all living things

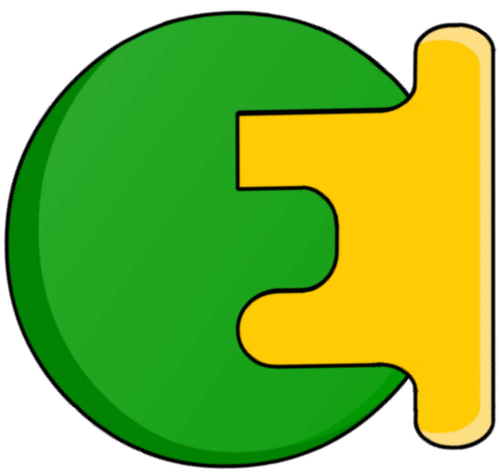
They are protein machines.

Lock and Key Hypothesis

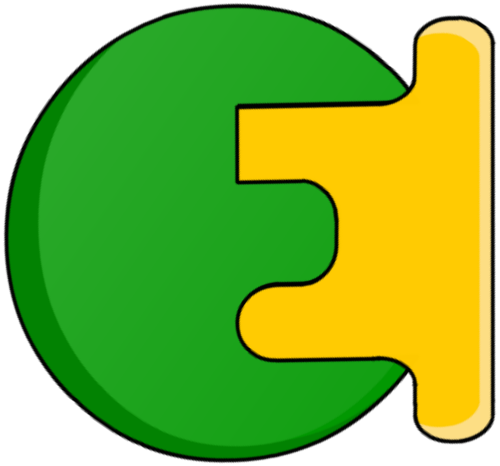
1. Enzymes have specific shapes for a **substrate** to attach.



2. When they **collide** the substrate fits the the **active site** of an enzyme **like a lock and key** and a **reaction occurs.**

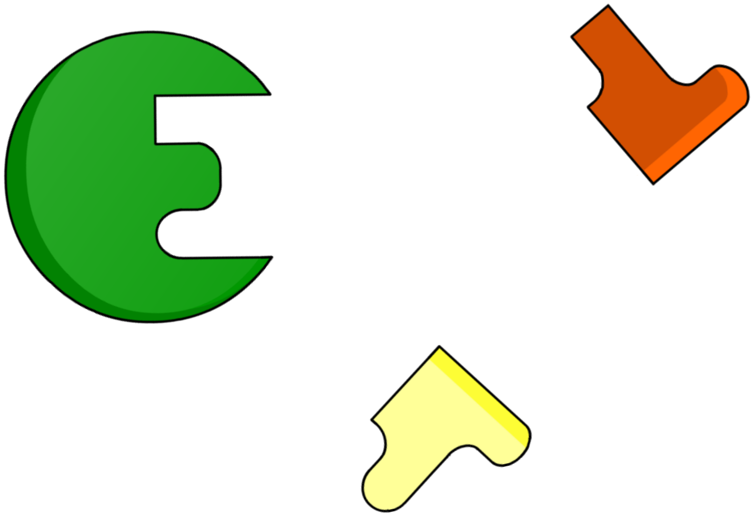


Lock and Key Hypothesis



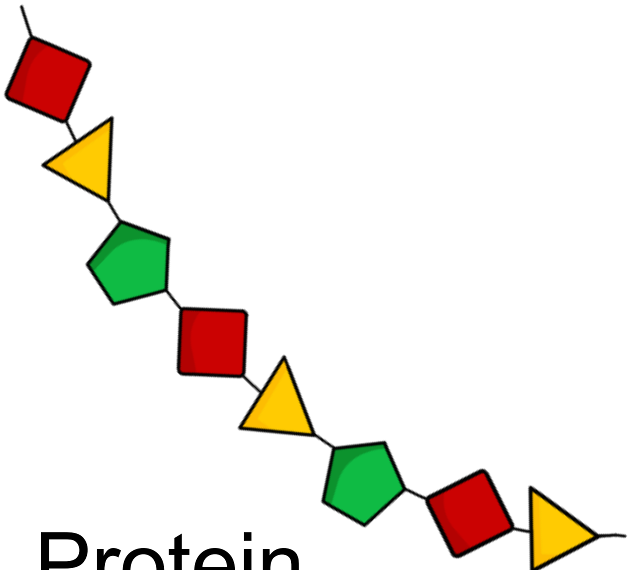
3. Once this happens we say the enzyme has **catalysed** the reaction which means, 'speeding it up'.

4. The substrate is broken up and enzyme is free to be used again.



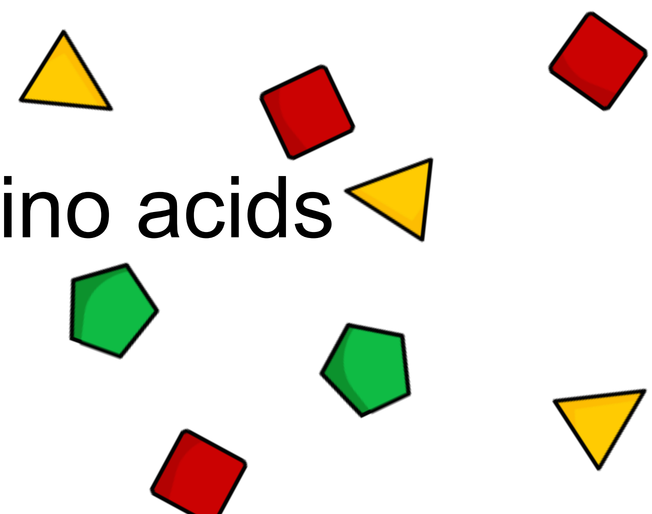
Lets verbally explain the process using protein as an example.

Keywords: Substrate, active site, specific, lock and key, reaction,

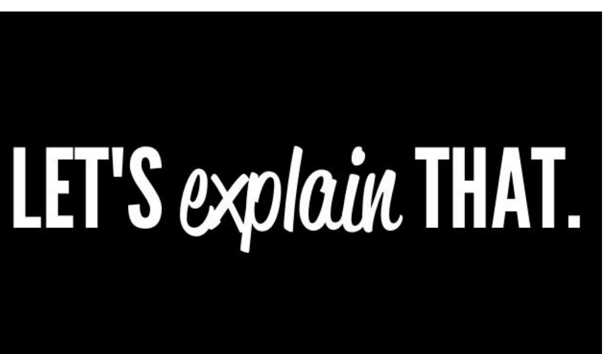


Protein

Pepsin enzyme

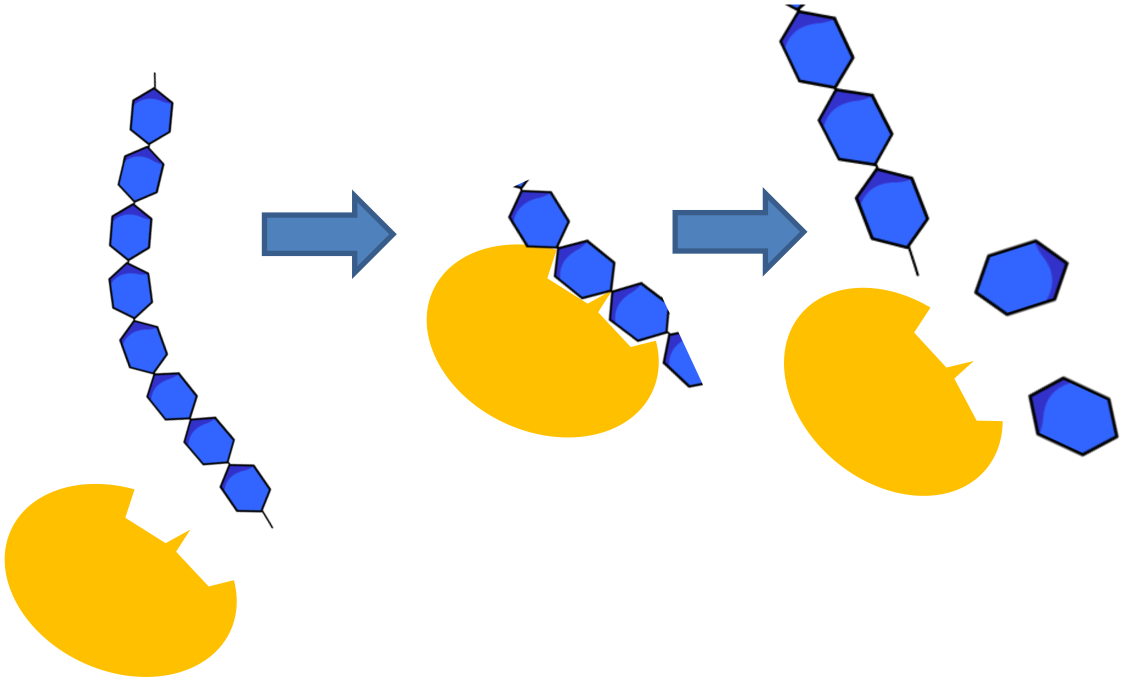


Amino acids



Activity: Complete your sheet to describe how amylase catalyses the reaction of carbohydrates.

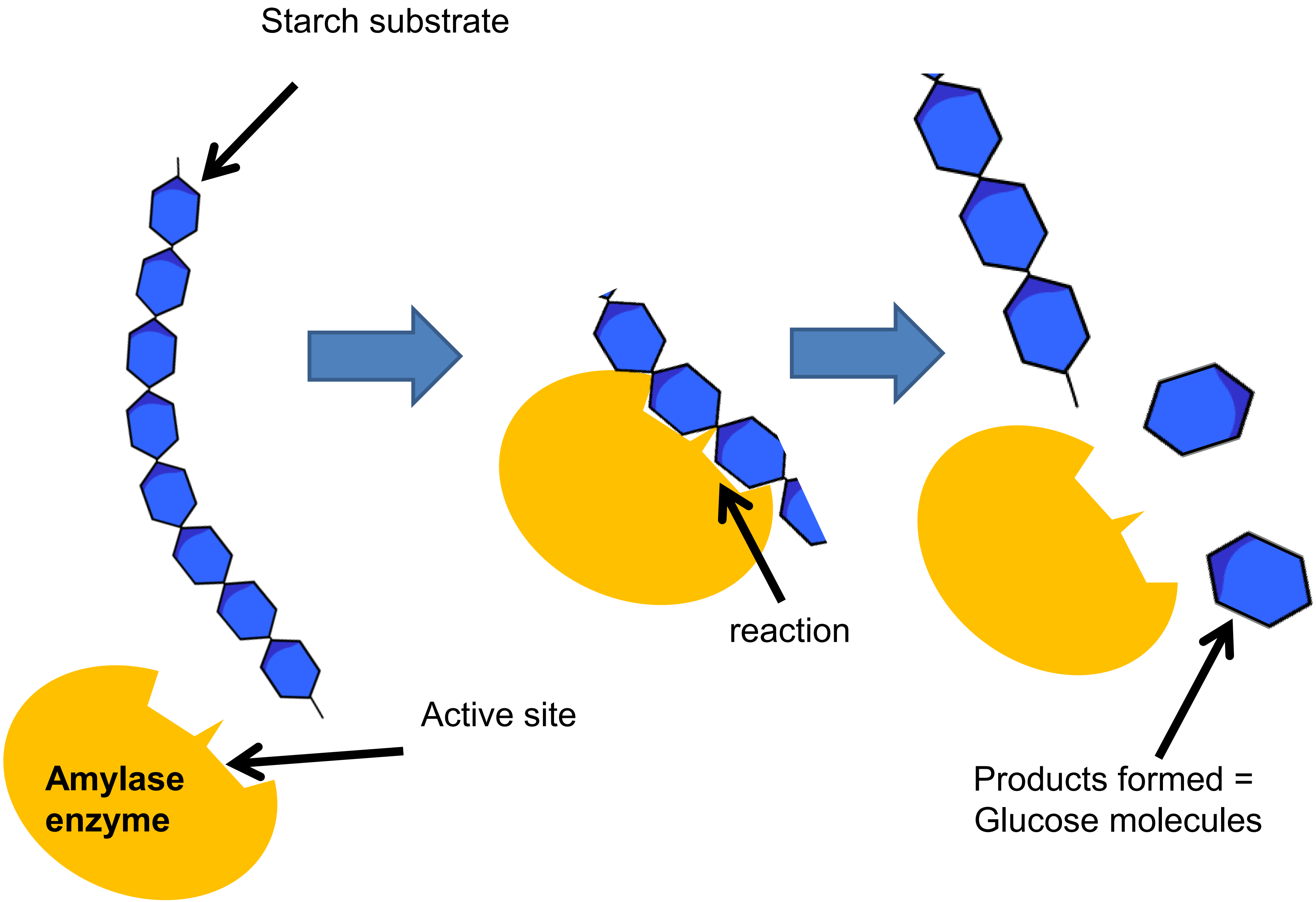
1. Must have labelled diagram
2. Must use all the key words



Keywords

Enzyme
Substrate
Product
Lock and key
Active site
Specific
Re-useable
Catalyse
Amylase
Glucose

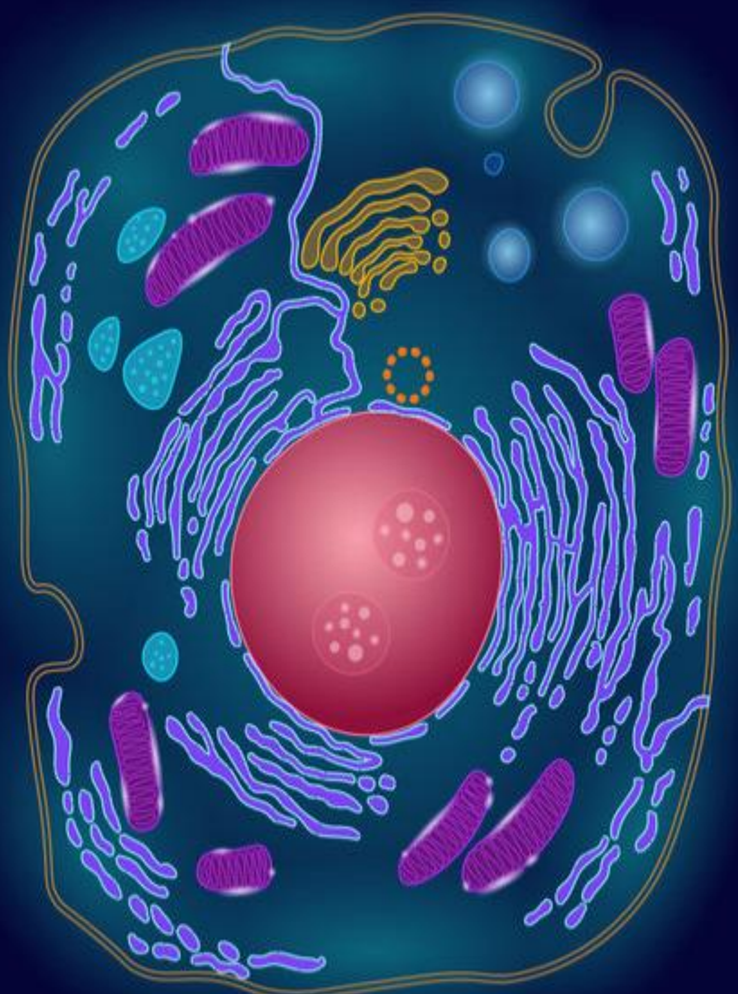
Review


Starch substrate
reaction
Active site
Amylase enzyme
Products formed = Glucose molecules

Review

The enzyme amylase has a specific shape to the substrate starch. The substrate binds to the active site like a lock and key. The enzyme connected to the substrate and catalyses a reaction. The bonds between the individual molecules in starch break and you are left with separated glucose molecules. The enzyme is not used up in the reaction so it is re-useable.

Enzymes affect rate of a reaction in a cell so enzymes link to **metabolism** of a cell.



REACTION
REACTION
REACTION
REACTION
\+ REACTION

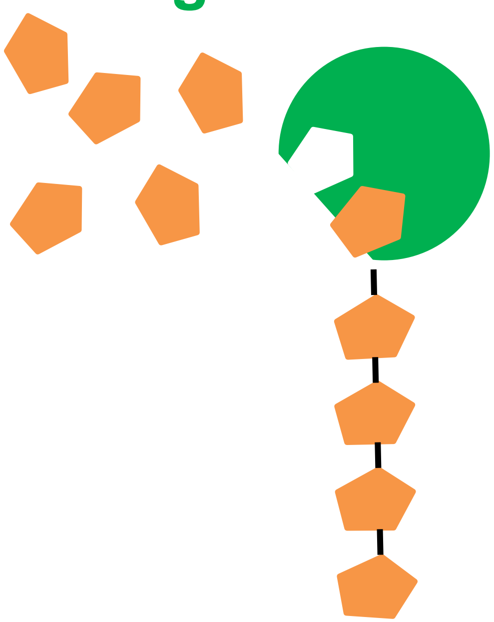
Metabolism

Metabolism is the sum of all reactions that happen inside of a cell or organism.

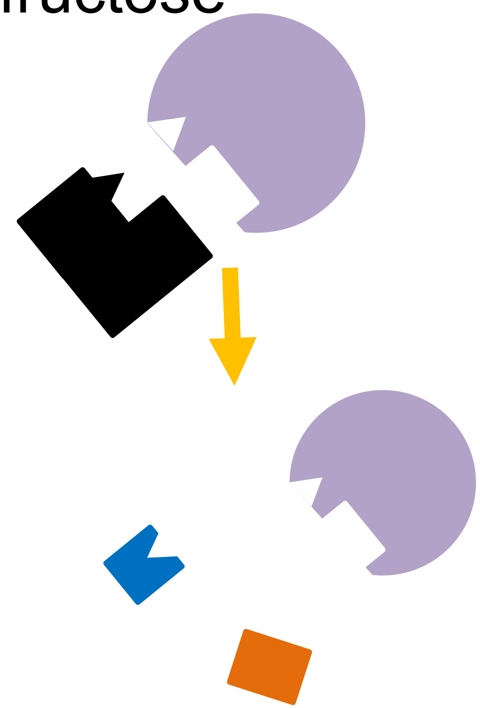
Different metabolic reactions

1	2	3
Building molecules from smaller ones	Changing a molecule into another	Breaking down one molecules into other smaller molecules

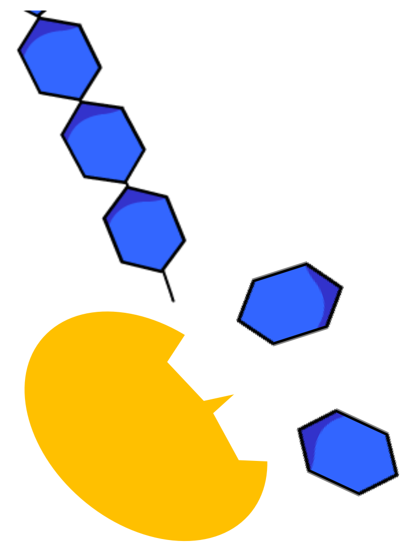
E.g. building **starch**, **cellulose** or **glycogen from glucose**



E.g. Glucose into fructose



E.g. Breaking down **large carbohydrate** molecule into **smaller glucose molecules.**

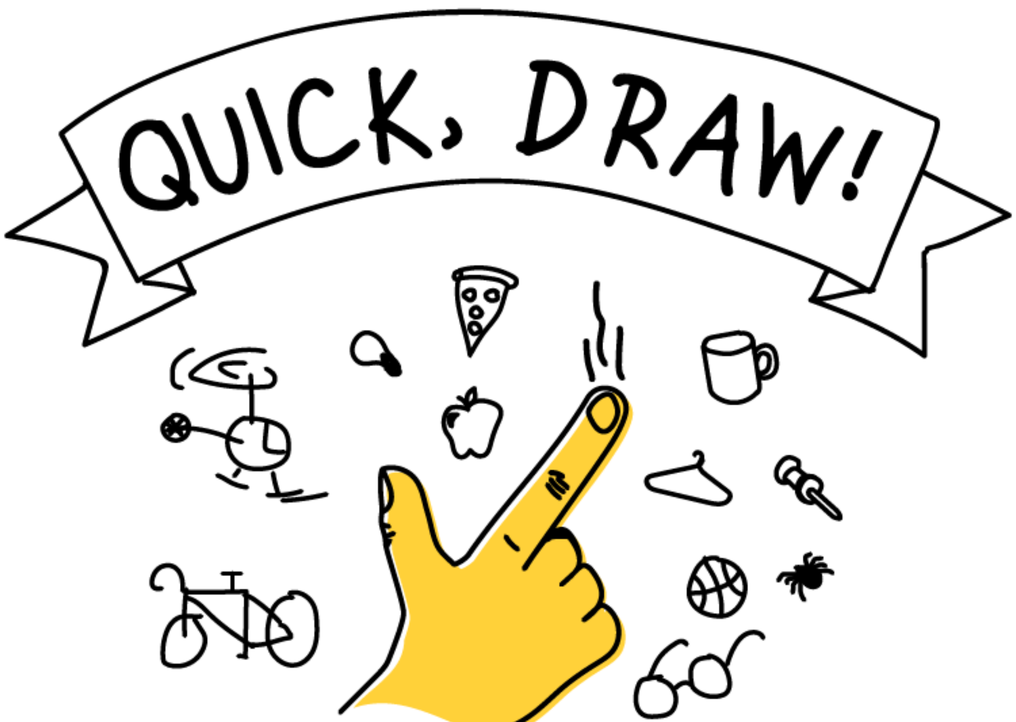


Activity: Show with a diagram the three different metabolic reactions.

1 **Building**

2 **Changing**

3 **Breaking Down**

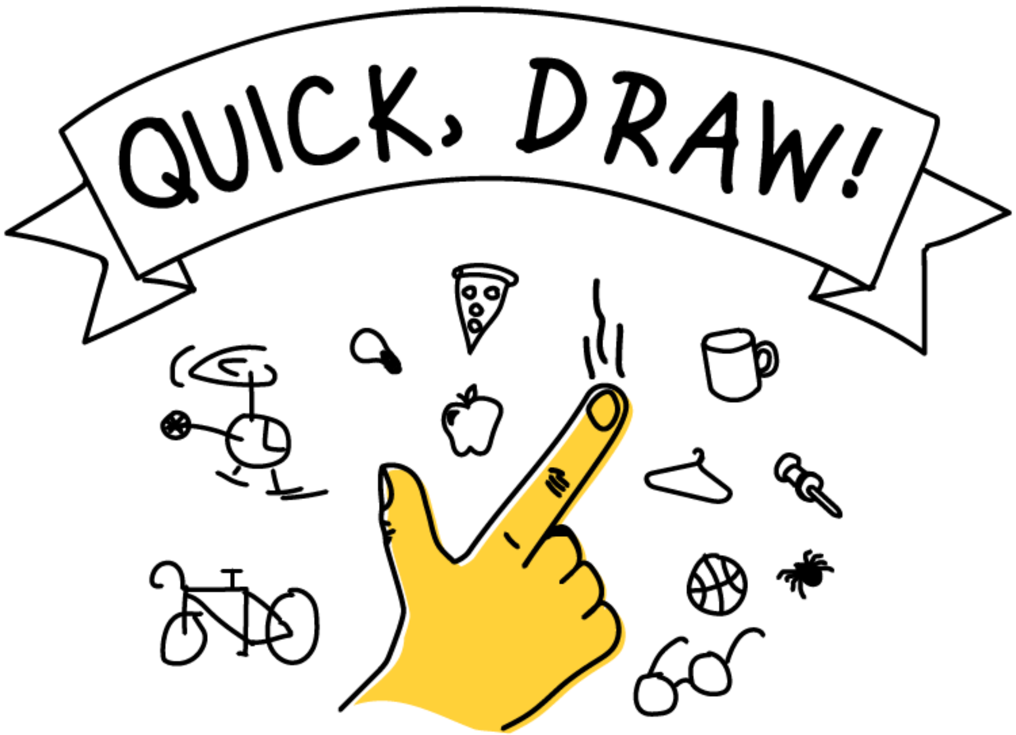


Review

1 **Building**

2 **Changing**

3 **Breaking Down**



LOb: Understand the how enzymes help digestion.

Keywords: Substrate, product, active site, denature, pH, lock and key, enzyme, catalyst

Success Criteria:

- ✓ Recall digestive enzymes; the products they form, where they are made, the substrates they act on
- ✓ Explain the lock and key hypothesis

Did you know… most enzymes are named with 'ase' at the end. However, enzymes discovered longer ago are not always with an 'ase' e.g. Pepsin or trypsin.

keyfacts

Enzymes facts 1) They are biological **catalyst** 2) They are made of **protein** 3) **re-useable** 4) **specific**

Digestive enzymes: **Amylase** → mouth → Carbohydrates **Pepsin** → stomach → protein **Lipase** → pancreas → Fat

Lock and key theory: **Substrate** attaches to enzyme **active site** and a **reaction** is catalysed producing a product.

Metabolism is the sum of all reactions in a cell or organism. cell reactions involve **building, changing** and **breaking down.**

Plenary: Answer these questions to assess your learning

1. Write three important enzyme facts

2. Describe three types of metabolic reactions

3. What do you know about the active site?

4. Explain what a substrate is and give an example

5. Unscramble these words taaytscl and iveistecta

Review

1. Write three important enzymes facts

Amylase, Pepsin (protease), Lipase

2. Describe three types of metabolic reactions.

Building up → Cellulose from glucose
Changing → Glucose into fructose
Breaking down → Starch into glucose

3. What do you know about the active site?

It is where the reaction occurs. It is specific in shape. Different for every enzyme.

4. Explain what a substrate is and give an example of a product made from a substrate

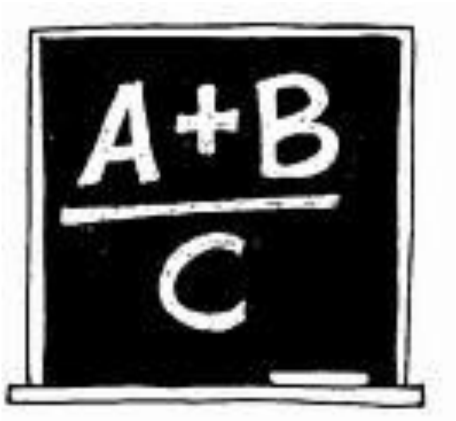
Substrate is something that reacts with an enzyme at the active Site. E.g. starch → glucose or protein → amino acid

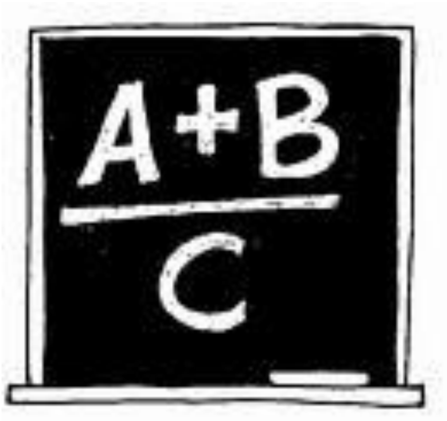
5. Unscramble these words CATALYST and ACTIVE SITE

Plenary

You can't get more frank than an equation
Create an equation summarising todays lesson
e.g

Eggs + flour + milk + sugar X oven = cake





Plenary: Exam Question

(c) Explain how the lock and key hypothesis models how enzymes work.

You may use labelled diagrams in your answer.

(3)

Question Number	Answer	Acceptable answers	Mark
6(c)	A explanation linking three of the following: • (enzyme and substrate have) complementary shapes • substrate fits into enzyme / enzyme substrate complex formed • reference to <u>active site</u> • enzymes break (chemical) bonds / form chemical bonds / (causes) reaction to occur / make products • Idea of products leaving enzyme (so that enzyme can be used again)	this may be awarded if clearly shown in an unlabelled diagram reject if active site is part of substrate	**(3)**